Ericsson PE Node Troubleshooting SOP - IND-PE-03: Indiranagar Exchange Routing Protocol Flap Issue

Node: IND-PE-03 | **Vendor:** Ericsson | **Severity:** Critical | **Alarm Code:** ALARM_6394

Description

This Standard Operating Procedure (SOP) outlines the necessary steps to troubleshoot and resolve the BGP route churn, OSPF adjacency loss, and ISIS LSP flooding issues affecting the Ericsson PE node, IND-PE-03, located at the Indiranagar Exchange. The failure is attributed to the repeated occurrence of routing protocol flap, which can lead to network instability, packet loss, and increased latency. This phenomenon is primarily caused by the dynamic nature of routing protocols such as BGP, OSPF, and ISIS. In the Ericsson equipment, the PE node's control plane is responsible for maintaining routing table information, while the data plane forwards packets based on the routing table. The routing protocol flap can be triggered by various factors, including configuration inconsistencies, hardware failures, or misbehaving neighbors. To address this issue, we will perform a series of diagnostic commands to identify the root cause and apply resolution steps to rectify the problem.

Pre-requisite Checks

- Verify the Indiranagar Exchange site is accessible and the Ericsson PE node, IND-PE-03, is online.

- Check the redundant link status to ensure no link failures are contributing to the issue.

- Confirm the Ericsson equipment software version is up-to-date and compatible with the current configuration.

Potential Root Causes

Cause	Probability
BGP route churn due to misconfigured neighbor IP addresses	High
OSPF adjacency loss resulting from incorrect interface configuration	Medium
ISIS LSP flooding caused by a malfunctioning router component	Low

Diagnostic Commands & Expected Outputs

```
Command: show bgp neighbors, Expected Output: Neighbor IP address: 10.1.1.1, State: Established, AS: 64512, Description: This command displays the BGP neighbor information, including the neighbor IP address, state, and AS number. The expected output indicates a stable BGP neighbor connection.

Command: show ospf interface, Expected Output: Interface IP address: 10.2.2.2, State: Up, Cost: 10, Description: This command displays the OSPF interface information, including the interface IP address, state, and cost. The expected output indicates a stable OSPF interface connection.
```

```
Command: show isis lsp, Expected Output: LSP ID: 123456, Sequence Number: 100, Metric: 10, Description: This command displays the ISIS LSP information, including the LSP ID, sequence number, and metric. The expected output indicates a stable ISIS LSP connection.
```

Resolution Steps

Step 1: Step: Verify BGP neighbor IP addresses, Command: show bgp neighbors, Description: Verify the BGP neighbor IP addresses are correct and stable. If incorrect, update the neighbor IP addresses to the correct values.

Step 2: Step: Check OSPF interface configuration, Command: show ospf interface, Description: Verify the OSPF interface configuration is correct and stable. If incorrect, update the OSPF interface configuration to the correct values.

Step 3: Step: Replace malfunctioning router component, Command: replace component /router, Description: If the ISIS LSP flooding is caused by a malfunctioning router component, replace the component with a new one.

Post-Check Verification

```
Command: show bgp neighbors, Expected Output: Neighbor IP address: 10.1.1.1, State: Established, AS: 64512, Description: Verify the BGP neighbor information is stable and correct.

Command: show ospf interface, Expected Output: Interface IP address: 10.2.2.2, State: Up, Cost: 10, Description: Verify the OSPF interface information is stable and correct.

Command: show isis lsp, Expected Output: LSP ID: 123456, Sequence Number: 100, Metric: 10, Description: Verify the ISIS LSP information is stable and correct.
```

Roll-back Procedures

- Step: Restore previous configuration, Command: load configuration /previous, Description: If the resolution steps fail, restore the previous configuration to the original state.

- Step: Restart the Ericsson PE node, Command: restart node /pe, Description: If the resolution steps fail, restart the Ericsson PE node to reset the control plane and data plane.